

Assignment

Interactive Dashboard

Tableau Practical File Link: - [Interactive_Dashboard](#)

Q1: Explain the difference between a Tableau Dashboard and a Tableau Story Point. Give two healthcare-related use cases.

- **Tableau Dashboard:** A dashboard is a collection of several related views (charts, maps, KPIs) displayed on a single screen. Its primary purpose is to provide an interactive, real-time bird's-eye view of data, allowing users to apply filters and drill down into the details on their own.
- **Tableau Story Point:** A story point is a sequence of visualizations (dashboards or worksheets) arranged in a specific narrative order, much like a presentation slide deck. Its primary purpose is to guide an audience through a structured data-driven narrative to explain insights or propose a business solution.

Healthcare Use Cases:

1. **Dashboard Use Case:** An *Emergency Room Live Monitor Dashboard* used by hospital staff to track the current number of admitted patients, available beds, and active on-duty doctors in real-time.
2. **Story Point Use Case:** An *Annual Budget and Hospital Performance Review Presentation* given to board members to walk them through how patient volumes changed over the year, which departments exceeded budgets, and where strategic investments should be made next.

Q2: What are Actions in Tableau? Explain how Filter Action can help hospital management analyse patient data.

- **Actions in Tableau:** Actions add context-driven interactivity to your worksheets and dashboards. They allow user interactions—such as clicking (Select), hovering, or using a menu—on one chart to trigger responses in other charts on the dashboard.
- **How Filter Action Helps Management:**

A Filter Action allows management to click on a specific data point in one chart to instantly filter all other visualizations on the dashboard.

- *Example:* If a manager clicks on the "Critical" slice of a Patient Outcome pie chart, a Filter Action will instantly update the rest of the dashboard to show *only* the critical patients, their respective hospital regions, and the specific doctors treating them. This helps management drill down into high-priority segments without creating dozens of separate reports.

Q3: Patient Distribution (Bar Chart)

Goal: Create a Bar Chart showing Total Patients by Hospital Region and add it to a Story Point.

- Step 1: In a new worksheet, drag the Hospital Region dimension to the Columns Shelf.
- Step 2: Drag Patient ID (or any unique patient identifier) to the Rows Shelf.
- Step 3: Right-click on the Patient ID pill in the Rows shelf, navigate to Measure, and select Count (or Count Distinct) to convert it into a bar chart showing patient volume.
- Step 4: Rename the worksheet to *"Patient Distribution"*.
- Step 5: Create a New Story, drag this worksheet onto the canvas, and add the caption: *"Patient Distribution by Region"*.

Q4: Department Performance (Customized Chart)

Goal: Create a visualization showing Treatment Cost by Department with labels and customized colours.

- Step 1: Drag the Department dimension to the Columns Shelf and the Treatment Cost measure to the Rows Shelf.
- Step 2: Drag Treatment Cost from your data pane and drop it onto the Label card in the Marks Card area to display the exact cost numbers above each bar.
- Step 3: Drag the Department dimension onto the Colour card.
- Step 4: Click on the Colour card, select Edit Colours, and choose a professional palette (such as *Classic 10* or *Teal*) to make each department easily distinguishable. Rename the sheet to *"Department Performance"*.

Q5: Treatment Trend Over Time (Line Chart)

Goal: Create a Line Chart of Treatment Cost vs. Admission Date filtered from March to September.

- Step 1: Drag Admission Date to the Columns Shelf. Right-click the pill and change it to continuous Month format (e.g., *May 2024*).
- Step 2: Drag Treatment Cost to the Rows Shelf. Tableau will automatically generate a Line Chart.
- Step 3: Drag Admission Date to the Filters Shelf.

- Step 4: Select Months in the filter setup window, then check only the boxes for March, April, May, June, July, August, and September, unchecking the rest. Click Apply. Rename the sheet to *"Treatment Trend Over Time"*.

Q6: Outcome Analysis (Pie Chart)

Goal: Create a Pie Chart showing Patient Outcome Distribution (Recovered / Critical / Under Treatment).

- Step 1: Create a new worksheet. In the Marks Card dropdown, change the chart type from *Automatic* to Pie.
- Step 2: Drag the Patient Outcome dimension onto both the Colour and Label cards.
- Step 3: Drag Patient ID onto the Angle card, right-click the pill, change its measure to Count, and drop another copy of it onto the Label card.
- Step 4: Right-click the Patient Count pill inside the Label card, select Quick Table Calculation, and choose Percent of Total so percentages appear on the pie chart slices. Rename it to *"Outcome Analysis"*.

Q7: Doctor Performance & Patient KPI Analysis

Task Part A - Doctor Performance (Ranked Bar Chart)

- Step 1: Drag Doctor Name to the Rows Shelf and Treatment Cost (Sum) to the Columns Shelf.
- Step 2: Click the Sort Descending icon on the top toolbar to automatically sort doctors from highest revenue generated to lowest.
- Step 3: *(Optional)* To strictly view top performers, drag Doctor Name to the Filters Shelf, go to the *Top* tab, select *By Field*, and set it to *Top 10 by Treatment Cost*. Rename the sheet to *"Doctor Performance"*.

Task Part B - KPI Creation (Total Patients Treated)

- Step 1: Open a blank sheet. Drag Patient ID straight onto the Text card in the Marks card area.
- Step 2: Right-click that pill and change its measure to Count.
- Step 3: Click on the Text card to format it: change the text font size to a large, bold format (e.g., *24pt Bold*) and add a label underneath reading *"Total Patients Treated"*. Rename the sheet to *"Total Patients KPI"*.

Q8: Interactive Dashboard

Goal: Build a cohesive, interactive dashboard containing the Region Chart, Department Cost Chart, and Outcome Chart with global filters.

- Step 1: Click the New Dashboard icon at the bottom toolbar.
- Step 2: Set the dashboard size to *Automatic* or *Desktop*. Drag the Region Chart (Q3), Department Cost Chart (Q4), and Outcome Chart (Q6) onto the dashboard layout.
- Step 3: Click on the Region Chart container, click the downward drop-down arrow in its top right corner, go to Filters, and select Hospital Region. Repeat the step to pull out the Department filter.
- Step 4: Click the drop-down arrow on the newly added filter boxes, go to Apply to Worksheets -> Selected Worksheets -> choose All using this data source. This ensures choosing a region updates all charts globally on the dashboard.

Q9: Storytelling Flow

Goal: Assemble a clean narrative story using at least 4 Story Points.

- Step 1: Click the New Story icon at the bottom of Tableau.
- Step 2: Story Point 1: Drag the *Patient Distribution (Q3)* sheet onto the canvas. Write the caption: "*1. Patient Distribution: Understanding Where Our Patients Come From*".
- Step 3: Click Blank to create a new point. Story Point 2: Drag the *Department Performance (Q4)* sheet onto the canvas. Write the caption: "*2. Department Cost Analysis: Pinpointing Highest Medical Expenses*".
- Step 4: Click Blank. Story Point 3: Drag the *Outcome Analysis (Q6)* sheet or your interactive dashboard onto the canvas. Write the caption: "*3. Outcome Analysis: Measuring Hospital Success Rates and Critical Cases*".
- Step 4: Click Blank. Story Point 4: Drag the *Doctor Performance (Q7 Part A)* sheet onto the canvas. Write the caption: "*4. Doctor Performance: Tracking High-Value Caseloads by Physician*".

Q10: Based on your visuals, answer:

1. Which department generates the highest treatment cost?

- How to answer: Look at your Department Performance Bar Chart (Q4). Find the tallest bar. (*Typically, in healthcare datasets, this is either Cardiology, Neurology, or Intensive Care*). State that department's name clearly.

2. Which region has more critical patients?

- How to answer: Look at your Interactive Dashboard (Q8). Click on the "Critical" slice of your Outcome Pie Chart. Because of the Filter Action, your Region Bar Chart will update to show only critical patients. Identify which region shows the highest bar count.

3. What business decisions can hospital management take from these insights?

- Strategic Recommendations:
 - Resource Allocation: Allocate more budget, specialized equipment, and medical personnel to the department generating the highest treatment costs to optimize operational efficiency.
 - Regional Infrastructure Expansion: In the region identified with the highest volume of critical patients, management should establish specialized emergency response systems, ICU expansion, or direct fast-track ambulance routing.
 - Seasonal Stock and Staffing Management: Using the time-trend chart (Q5), management can see if patient expenses spike in specific months (e.g., during summer or monsoon seasons) and adjust clinical staffing and pharmaceutical inventories beforehand to reduce emergency overhead.